FlexTime Support Chatbot System Documentation

1. System Overview

The FlexTime Support Chatbot is designed to help customers with common support questions, classify their requests, generate professional responses, and escalate cases that require human review.

The chatbot follows a three-stage prompt chain:

1. **Stage 1 – Intent Classifier:** Identifies the customer's issue category and sentiment.
2. **Stage 2 – CARE Responder:** Generates an empathetic, professional response based on the customer's query and FlexTime's support policies.
3. **Stage 3 – Reflection and Verification:** Checks the draft response for policy violations, incorrect promises, inappropriate tone, and security risks. If a response fails verification, the system routes the case for escalation.

2. Target Personas

- **Customers:** People who need assistance with billing issues, account management, technical problems, or general inquiries.
- **Customer Support Agents:** Staff who use chatbot-generated responses to handle customer requests consistently.
- **Support Managers:** Staff who review escalated billing disputes, policy exceptions, and complex customer issues.

3. Chatbot System Instructions

Role

Act as a professional customer support representative for FlexTime. Respond clearly, politely, and empathetically while following the official support policy.

Intent Classification

Classify each customer query into one of these categories:

- Billing
- Technical Support
- Account Management
- General Inquiry
- Urgent Escalation

Classify customer sentiment as Satisfied, Neutral, Angry, or Frustrated.

Response Generation

- Acknowledge the customer's concern and apologize when appropriate.
- Use simple, clear language and maintain a professional tone.
- Keep responses under 100 words.
- Use only verified information from the support policy.
- Ask for required information when necessary.
- Never claim an action has been completed unless it has been confirmed.

Reflection and Verification

Before sending a response, check that it:

- Follows FlexTime's support policies.
- Uses an appropriate tone.
- Does not promise unauthorized refunds or unsupported outcomes.
- Does not disclose confidential information.
- Routes policy violations and cases requiring approval to human support.

4. Security Guardrails and Support Policies

Refund Policy

Support agents may issue refunds up to $20 without approval. Billing disputes involving refunds above $20 must be escalated to the appropriate team.

Account Management Policy

Customers must verify their current account ID and billing ZIP code before their profile email can be changed.

Privacy Guardrails

The chatbot must never disclose billing API keys, database hashes, or internally marked ticket numbers to customers.

Operating Hours

FlexTime support operates Monday through Friday, from 9:00 AM to 5:00 PM EST.

Human Escalation

Escalate refund requests above $20, flagged policy violations, and cases that require human review. Do not claim a refund has been approved or processed until that outcome is confirmed.

5. Prompt Chain and Logical Routing Diagram

The following Mermaid.js diagram illustrates how a customer query moves through classification, policy checks, response generation, verification, and escalation.

```
flowchart TD
    A["Customer Query"] --> B["Intent Classifier - Stage 1"]
    B --> C["Category: Billing / Account / Technical Support"]
    C --> D{"Check Policy Limits"}

    D -->|"Within 20limit/StandardQuery"|E["CAREResponder − Stage2"]D −−−> |"20 Limit Exceeded / Urgent Escalation"| H["Escalation Ticket Prompt"]

    E --> F["Draft Reply"]
    F --> G["Reflection and Verification - Stage 3"]

    G -->|"Approved"| I["Send Reply to Customer"]
    G -->|"Flagged Violation"| H

    H --> J["Human Support Queue"]

    classDef process fill:#e8e0ff,stroke:#8b6be8,color:#222
    classDef decision fill:#f2eaff,stroke:#8b6be8,color:#222
    classDef finalStep fill:#e5f4e8,stroke:#559566,color:#222

    class A,B,C,E,F,G,H process
    class D decision
    class I,J finalStep
```

6. Example Workflow

Scenario: A customer requests a $120 refund because their annual plan is not working.

1. **Stage 1:** Classifies the query as Billing and the sentiment as Frustrated.
2. **Policy check:** Detects that $120 exceeds the $20 refund limit.
3. **Stage 2:** Generates an empathetic response explaining that the request requires escalation.
4. **Stage 3:** Verifies that the response does not promise an unauthorized refund or disclose internal information.
5. **Escalation:** Generates a handover ticket for the human support team.
6. **Final outcome:** The customer receives an appropriate update, and the refund remains pending review until a decision is confirmed.

7. Expected Benefits

- Consistent and professional customer communication.
- Faster classification and handling of common queries.
- Improved compliance with refund and account security policies.
- Reduced risk of unauthorized promises and confidential data leaks.
- Clear escalation paths for complex or policy-sensitive cases.

8. Conclusion

The FlexTime Support Chatbot combines intent classification, policy-based response generation, and reflection-based verification to provide reliable customer support. Its escalation process ensures that requests beyond the chatbot's authority are directed to human support instead of receiving unapproved resolutions.